Dual-Attention BiGRU for Human Activity Recognition

Updated Project Plan — Literature Review, Gap Analysis, Methodology, Architecture & Timeline

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1. Background

Human Activity Recognition (HAR) from wearable/inertial sensor data is a time-series classification problem that maps naturally onto RNN (LSTM/GRU) and attention architectures without requiring CNNs. It has multiple free, well-benchmarked public datasets (UCI-HAR, WISDM, PAMAP2) and models that train in minutes-to-hours on a single GPU — realistic for a solo 3-month project.

This document supersedes the original proposal after a fresh literature check (July 2026) revealed that several recent papers had already implemented the originally-proposed idea. The project has been repositioned around a gap that remains genuinely open.

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2. Literature Review

2.1 Original 10-paper review (foundation papers)

#	Topic	Algorithm	Dataset	Key Contribution	Limitation / Gap
1	Attention-based RNN with information-gain sensor placement	BiLSTM + Attention	UCI Opportunity Challenge	Sensor-selection + attention re-weighting	Dataset-specific, no cross-dataset test, no smartphone-only setup
2	Attention-based Residual BiLSTM	Residual BiLSTM + Attention	UCI-HAR, WISDM, KU-HAR	Fixes confusion between similar activities (walking vs. walking-upstairs)	No cross-subject/cross-dataset transfer study
3	GRU-based Hierarchical Attention	GRU + Hierarchical Attention	Nurse-activity dataset	Handles class imbalance	Cross-subject accuracy much lower (~66%); small, non-public dataset
4	Lightweight Attention-GRU for WiFi CSI HAR	GRU + Self-Attention + pruning	WiFi CSI (ARIL etc.)	252M → 0.058M params, ~98.9% accuracy	Only RF/CSI signals, not IMU; unclear if pruning transfers
5	GRU with Dual Attention (channel + temporal)	BiGRU + Dual Attention	Custom multi-sensor dataset	Splits "where to look" (channel) from "when to look" (temporal)	Not tested on UCI-HAR/WISDM/PAMAP2; no ablation isolating each attention type
6	Transformer for HAR (no RNN/CNN)	Vanilla Transformer Encoder	Smartphone accel+gyro	Attention alone matches CNN+LSTM	Generic positional encoding, no data-efficiency analysis
7	CNN-LSTM with Self-Attention	CNN + LSTM + Self-Attention	H-Activity, MHEALTH, UCI-HAR	99.93% / 98.76% / 93.11%	Gap between own data (99.9%) and UCI-HAR (93%) suggests overfitting

#	Topic	Algorithm	Dataset	Key Contribution	Limitation / Gap
8	Multi-branch CNN-BiLSTM	CNN + BiLSTM	Multiple public datasets	Strong cited baseline	Reference-only; CNN-containing "gold standard"
9	CNN-BiLSTM Parallel + Attention (noisy data)	CNN + BiLSTM + Attention	UCI-HAR, WISDM	Robust to noisy/missing sensor data	Unclear if pure RNN+Attention can match this without CNN
10	Multi-Agent Attention-CNN-BiLSTM (ensemble)	CNN + BiLSTM + Attention + ML	Disability-assistance dataset	Ensemble accuracy gains	Deployment complexity, no latency/resource analysis

Pattern: the strongest recent baselines (rows 7–10) are CNN+RNN hybrids; pure RNN+Attention papers (rows 1, 2, 3, 5, 6) are fewer, mostly single-dataset, and rarely report cross-dataset generalization.

2.2 Newly identified papers (found on re-search, July 2026) — critical addition

These were **not** in the original review and directly overlap with the original proposal's claimed novelty:

Paper	Venue / Date	Architecture	Datasets	Result
Dual-INABA (Dual Inception-Attention-BiGRU-Attention)	*Scientific Reports*, Dec 2025	Inception + Attention + BiGRU + Attention, wavelet-transform preprocessing	PAMAP2, UCI-HAR, WISDM + self-collected WSMC-HAR	98.15% / 98.89% / 96.78% accuracy; ablations isolate BiGRU vs. Inception contribution
TGA-HAR	*Sensors (MDPI)*, Sept 2025	TCN + BiGRU + Attention	WISDM, USC-HAD, PAMAP2, Real-World IDLab	Robust generalization across sensor configurations
MHAT-Net	*MDPI*, April 2026	Parallel BiGRU + Transformer dual-pathway	Smartphone inertial (motionless activities)	Targets weak-signature stationary activities
CNN-BiGRU-Transformer	*Applied Sciences*, Nov 2025	Multi-branch CNN + BiGRU + Transformer	Wearable sensor data	Improved long-term temporal dependency capture
AttnFusionGRU	*ScienceDirect*, Sept 2025	Dual-branch attention + SqueezeNet + BiGRU	UCF101, Human3.6M	Spatial + temporal dual attention (vision-based, not IMU)

Cross-dataset transfer (originally proposed as the "key novelty result" in Weeks 10–11) is also a mature, active subfield with methods well beyond simple fine-tuning: Deep CORAL domain adaptation, self-supervised cross-domain pretraining (UniMotion, CrossHAR), and data-mixture optimization (HAR-DoReMi, ~6.5% average accuracy improvement over prior state-of-the-art). Naive train-then-fine-tune is no longer a novel contribution in this space.

3. Gaps Identified (Updated)

Gap	Status in literature	Still open?
Dual attention (channel + temporal) + BiGRU backbone	**Closed** — Dual-INABA (Dec 2025) does this on the exact 3 benchmark datasets	❌ No
Cross-dataset transfer with fine-tuning as sole novelty	**Closed / surpassed** — CORAL, CrossHAR, HAR-DoReMi far more sophisticated	❌ No
CNN-free (pure RNN+Attention) model compared against CNN-hybrids on **params, FLOPs, and inference latency**, not just accuracy	Not directly reported by Dual-INABA, TGA-HAR, or CNN-BiGRU-Transformer — these report accuracy only	✅ **Yes**
Evaluation using **subject-independent (leave-subjects-out) cross-validation**, multiple seeds, and statistical significance testing across all 3 benchmark datasets	Most reviewed papers (including new ones) report single random-split accuracy; no paired significance testing found	✅ **Yes**

Conclusion: the project's novelty is repositioned from *architecture novelty* ("we combined X and Y first") to *methodological novelty* ("we are the first to report this specific efficiency-and-rigor combination for a CNN-free dual-attention BiGRU on these three benchmarks"). This is a defensible, currently-unclaimed gap.

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4. Final Novelty Statement

> Recent HAR literature (Dual-INABA, TGA-HAR, MHAT-Net, CNN-BiGRU-Transformer) reports accuracy-only comparisons on single train/test splits. No reviewed CNN-free BiGRU + dual-attention model reports parameter/FLOPs/latency accounting alongside subject-independent, multi-seed, statistically validated evaluation across UCI-HAR, WISDM, and PAMAP2 — this project addresses that combined efficiency-and-rigor gap.

The two novelty pillars

1. **Efficiency accounting** — Dual-Attention BiGRU vs. a CNN+BiGRU reference baseline and other baselines, compared on **accuracy, params, FLOPs, and inference latency**, all reimplemented under identical conditions (not copied from literature tables).
2. **Evaluation rigor — Subject-independent (LOSO) cross-validation**, ⩾3 seeds per fold, mean±std reporting, and **paired statistical significance testing** (Wilcoxon signed-rank) — instead of a single random-split accuracy number.

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5. Updated Methodology

5.1 Models to implement (4 total)

#	Model	Role	File
1	Plain BiGRU	Baseline 1	`src/models/bigru_baseline.py`
2	BiLSTM + single attention	Baseline 2 (reproduces literature rows 1/2)	`src/models/bilstm_attention_baseline.py`
3	Small CNN + BiGRU (2–3 conv layers)	**New** — reference "heavy" baseline; required to make an efficiency claim credible	`src/models/cnn_bigru_reference.py`
4	Dual-Attention BiGRU (channel + temporal attention)	Proposed model	`src/models/dual_attention_bigru.py`

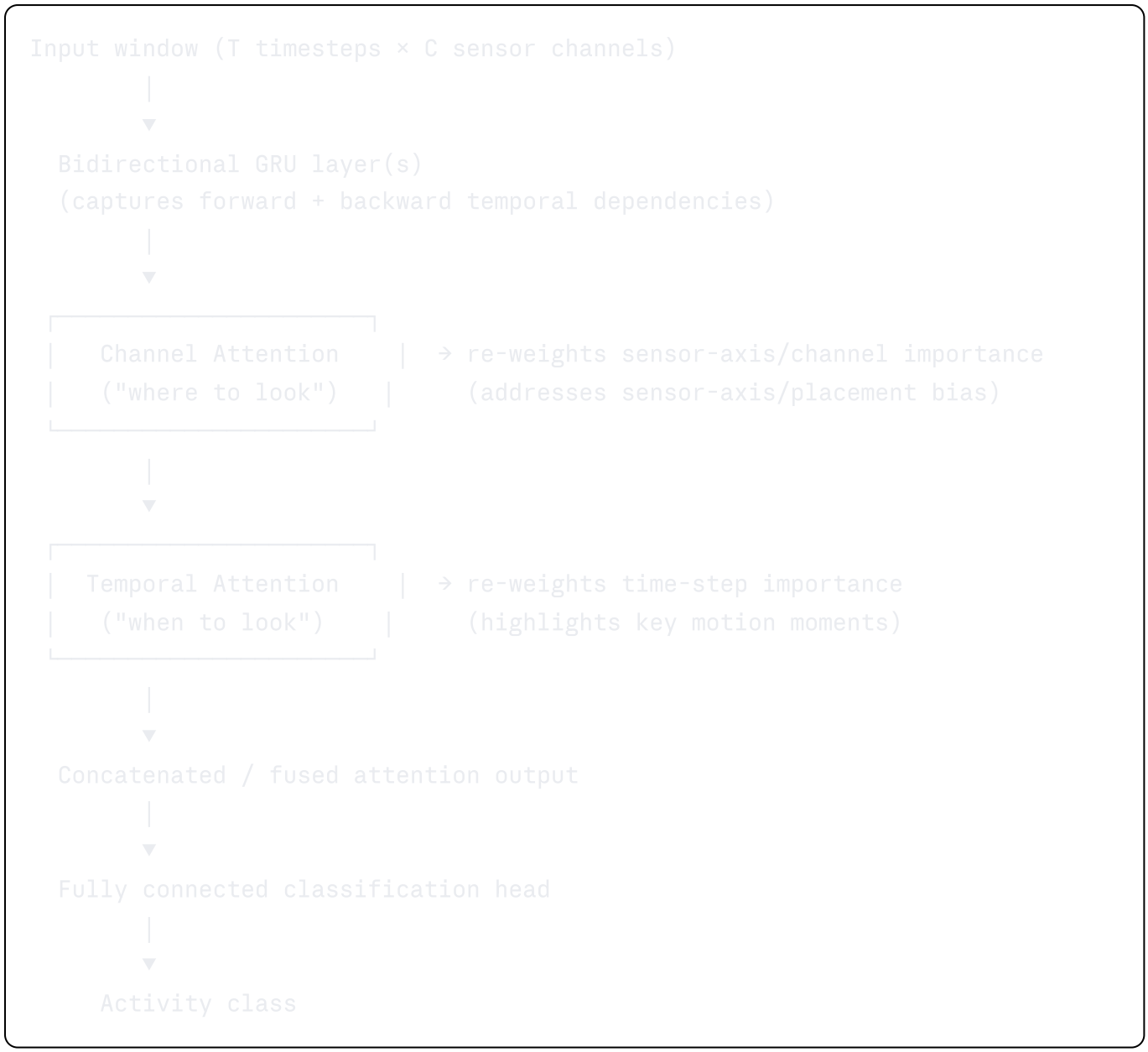
5.2 Evaluation protocol

- **Two split modes**, both reported:
  - Official train/test split (as used in most literature, for comparability).
  - **Subject-independent k-fold (leave-N-subjects-out)** — the headline result.
- **≥3 random seeds** per model per fold; report mean ± standard deviation.
- **Efficiency metrics** logged for every model: parameter count, FLOPs (via `thop`/`ptflops`), and inference latency (ms/sample, batch=1 and batch=32, CPU and GPU if available).
- **Statistical testing**: Wilcoxon signed-rank test (or paired t-test) comparing Dual-Attention BiGRU's per-fold scores against each baseline's per-fold scores, with reported p-values and effect size.
- **Master comparison table** (`results/tables/master_comparison.csv`) — columns: model, dataset, mean accuracy ± std, mean F1 ± std, params, FLOPs, latency (ms), p-value vs. Dual-Attention BiGRU.
- **Optional bonus** (only if time remains after Week 11): a light cross-dataset transfer experiment (train on one dataset, fine-tune on another) — framed as a supplementary result, not a core claim.

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6. Architecture

6.1 Dual-Attention BiGRU (proposed model)



- **No CNN or Transformer branch** — pure recurrent + attention design, kept lightweight by construction.
- Channel attention and temporal attention are implemented as separate, ablatable modules in `attention_modules.py` so their individual contributions can be isolated (addresses the ablation gap flagged in literature row 5).

6.2 Reference baselines

- **BiGRU baseline**: BiGRU → FC head (no attention).
- **BiLSTM + single attention**: BiLSTM → one attention layer → FC head (reproduces literature rows 1/2 approach).
- **CNN + BiGRU reference**: 2–3 Conv1D + pooling layers → BiGRU → FC head (the "heavy" model the efficiency claim is measured against).

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7. Datasets

Dataset	Subjects	Sensors	Sampling Rate	Activities	Source
UCI-HAR	30	Waist-mounted smartphone accelerometer + gyroscope	~50 Hz	6 (walking, upstairs, downstairs, sitting, standing, laying)	UCI Machine Learning Repository (`archive.ics.uci.edu`)
WISDM	36	Phone/watch accelerometer (+ gyroscope in newer release)	~20 Hz	6 (walking, jogging, upstairs, downstairs, sitting, standing)	WISDM Lab (Fordham University) / Kaggle mirrors
PAMAP2	9	Chest, hand, ankle IMUs + heart rate	100 Hz	12+ (protocol + optional activities)	UCI Machine Learning Repository

Harmonization requirement: the three datasets differ in sampling rate and sensor placement. `preprocessing.py` must implement explicit resampling and channel-alignment logic before any cross-dataset comparison is meaningful — this must be documented in the final report.

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8. Expected Outcomes

1. A complete, reproducible 4-model × 3-dataset comparison table (accuracy, macro-F1, params, FLOPs, latency) with no placeholder cells.
2. Subject-independent (LOSO) cross-validation results with mean ± std over ⩾3 seeds, for every model on every dataset.
3. At least one statistical significance test (Wilcoxon or paired t-test) with reported p-values, comparing Dual-Attention BiGRU against each baseline.
4. Evidence (or disproof) of the hypothesis that a CNN-free dual-attention BiGRU can match or approach CNN-hybrid accuracy at meaningfully lower parameter count / FLOPs / latency — the paper's headline result either way.
5. A related-work section explicitly naming and differentiating from Dual-INABA, TGA-HAR, MHAT-Net, CNN-BiGRU-Transformer, and AttnFusionGRU.
6. A public GitHub repository with weekly commit history matching the timeline below, serving as provenance evidence for the work.

7. (Stretch, optional) A supplementary cross-dataset transfer result, framed as exploratory rather than the core contribution.

Realistic target venues: MDPI *Sensors*, MDPI *Applied Sciences*, or IEEE *Access* (Scientific Reports is a harder sell now that Dual-INABA occupies very similar territory there). Student/national-level DL/ML conferences remain a solid fallback.

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9. Week-Wise Timeline (Updated)

Week	Deliverables	Notes / Changes from original plan
1	`docs/research_gap_notes.md`, `report/literature_review.md` (with new papers added), `README.md` draft, `requirements.txt`	Literature table finalized including Dual-INABA, TGA-HAR, MHAT-Net, CNN-BiGRU-Transformer, AttnFusionGRU with explicit differentiation
2	`data/download_data.py`, `data/raw/ucihar/` populated, `.gitignore`	UCI-HAR downloaded and readable; repo skeleton committed
3	`src/preprocessing.py`, `src/data_loader.py`, `data/processed/ucihar/`, `notebooks/01_data_exploration.ipynb`	**Add subject-wise k-fold (LOSO) split logic now**, alongside standard split
4	`src/models/bigru_baseline.py`, `src/train.py`, `src/utils.py`, `configs/ucihar_config.yaml`, `checkpoints/bigru_baseline/`	**Add params/FLOPs/latency logging to `evaluate.py` here** so it's reused for every later model
5	`src/models/bilstm_attention_baseline.py`, `checkpoints/bilstm_attention/`, first entry in `results/tables/`	Baseline 2 trained and compared against Baseline 1
6	**New:** `src/models/cnn_bigru_reference.py`, `checkpoints/cnn_bigru_reference/`	Added task — the "heavy" reference baseline required for the efficiency claim
7	`src/attention_modules.py` (channel + temporal attention complete), `src/models/dual_attention_bigru.py` (complete), `notebooks/04_dual_attention_bigru.ipynb`	Full model trained on UCI-HAR; **all 4 models now run with 3-seed × k-fold LOSO**; first significance test
8	`data/raw/wisdm/`, `data/processed/wisdm/`, `configs/wisdm_config.yaml`	All 4 models trained on WISDM; results added to `results/tables/`
9	`data/raw/pamap2/`, `data/processed/pamap2/`, `configs/pamap2_config.yaml`, `results/tables/master_comparison.csv`	All 4 models trained on PAMAP2; master comparison table (accuracy, F1, params, FLOPs, latency, p-values) completed

Week	Deliverables	Notes / Changes from original plan
10	Buffer / catch-up week; optional light cross-dataset transfer experiment if time allows	Deliberate safety margin — LOSO + multi-seed + stats-test infrastructure is the highest-risk timeline item
11	Finalize figures: accuracy-vs-efficiency scatter plot, LOSO boxplots per model, significance table, `results/figures/`	Key visual deliverables for the paper
12	`report/project_report_draft.docx`, `report/slides.pptx`, `README.md` finalized	Full report/paper draft, presentation, reproducible README

Commit to GitHub at the end of every week, even if incomplete.

10. Project Folder Structure

```
dual-attention-bigru-har/
│
├── README.md
├── requirements.txt
├── .gitignore
│
├── data/
│   ├── download_data.py
│   ├── raw/
│   │   ├── ucihar/
│   │   ├── wisdm/
│   │   └── pamap2/
│   └── processed/
│       ├── ucihar/
│       ├── wisdm/
│       └── pamap2/
│
├── src/
│   ├── __init__.py
│   ├── data_loader.py            # includes subject-wise LOSO split mode
│   ├── preprocessing.py          # includes cross-dataset harmonization
(resampling, channel alignment)
│   ├── attention_modules.py      # channel-attention & temporal-attention
(ablatable)
│   ├── models/
│   │   ├── __init__.py
│   │   ├── bigru_baseline.py
│   │   ├── bilstm_attention_baseline.py
│   │   ├── cnn_bigru_reference.py     # NEW — heavy reference baseline
│   │   └── dual_attention_bigru.py
│   ├── train.py                  # multi-seed training loop
│   ├── evaluate.py               # accuracy, F1, confusion matrix, params,
FLOPs, latency
│   ├── stats_test.py             # NEW — Wilcoxon / paired t-test
significance testing
│   ├── cross_dataset_transfer.py  # optional bonus experiment
│   └── utils.py
│
├── configs/
│   ├── ucihar_config.yaml
│   ├── wisdm_config.yaml
│   └── pamap2_config.yaml
│
├── notebooks/
```

```
|   ├── 01_data_exploration.ipynb
|   ├── 02_baseline_bigru.ipynb
|   ├── 03_baseline_bilstm_attention.ipynb
|   ├── 04_cnn_bigru_reference.ipynb      # NEW
|   ├── 05_dual_attention_bigru.ipynb
|   ├── 06_multi_dataset_runs.ipynb
|   └── 07_cross_dataset_transfer.ipynb   # optional bonus
|
├── checkpoints/
|   ├── bigru_baseline/
|   ├── bilstm_attention/
|   ├── cnn_bigru_reference/
|   └── dual_attention_bigru/
|
├── results/
|   ├── tables/                 # includes master_comparison.csv with
efficiency + p-values
|   ├── figures/                # accuracy-vs-efficiency plots, LOSO
boxplots, significance table
|   └── logs/
|
├── report/
|   ├── literature_review.md    # updated with 5 new papers + explicit
differentiation
|   ├── project_report_draft.docx
|   └── slides.pptx
|
└── docs/
    └── research_gap_notes.md   # updated final novelty statement
```

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11. Minimum Bar for Submission (Non-Negotiable)

By end of Week 12, the following must be true — not partial:

1. Full 4×3 comparison table (4 models × 3 datasets), every cell filled.
2. Subject-independent CV results with mean ± std over ⩾3 seeds per fold.
3. At least one statistical significance test with reported p-values.
4. Related-work section explicitly differentiating from the 5 newly identified papers.
5. Public GitHub repo with weekly commit history matching this timeline.
6. Manuscript formatted for the target journal (MDPI Sensors/Applied Sciences template, IEEE Access as backup).

Items 1–4 are the actual research contribution — if time runs short, protect these first. Items 5–6 are packaging and can be finished last.